

GravityPro — Third-Party APIs & Services (Paid vs Free + Limits)

This documents every external API/service the **website (landing-react)**, **mobile app**, and **backend** integrate with — whether it's **paid or free**, and for free ones the **monthly limit / free tier**.

⚠ Pricing & free-tier limits change often. Numbers below are typical/approximate (2024–2025). Always confirm on each provider's current pricing page before launch. "Free integration" for payment gateways means no monthly fee — they take a % **per transaction**.

Evidence: derived from `backend/package.json` + `.env.example`, `mobile/package.json`, `landing-react/package.json`, and code references found in `backend/src`, `mobile/src`, `landing-react/src`.

♥ Cost-Optimized Stack (what we changed to save money)

GravityPro now runs a **free-first** architecture. The swaps we made:

- **Google Maps → free Leaflet + OpenStreetMap + CARTO basemaps** (and Esri ArcGIS for the satellite layer). No API key, no billing card, no monthly map bill.
- **SMS-first → Email-first OTP.** Email OTP (free SMTP tiers / dev mode) is now the **primary** verification method. SMS OTP (MSG91) is **optional and off by default**, so the main recurring paid cost is avoided.
- **Cloudflare R2 kept** for storage — generous free tier and **\$0 egress**.

Practical effect: With this setup the app can run with **ZERO mandatory monthly API costs** — you only pay payment-gateway %-fees when collecting money, plus your own server/DB/storage as usage grows. **SMS is the only optional paid add-on** (enable MSG91 only if you want phone-OTP).

🔑 QUICK SUMMARY TABLE

#	Service	Category	Paid / Free	Free monthly limit (approx)	Used in
1	Neon PostgreSQL	Database	Free tier + Paid	~0.5 GB storage, ~190 compute-hrs/mo	Backend
2	Cloudflare R2	File/object storage	Free tier + Paid	10 GB storage, 1M writes, 10M reads/mo, \$0 egress	Backend (avatars/media)
3	Google Maps Platform	Native maps	REMOVED — replaced by free OSM/Leaflet/CARTO	n/a (no longer used)	— (not used)
4	OpenStreetMap tiles	Map tiles	FREE	Fair-use policy (no bulk/high-volume)	Mobile + Web (Leaflet)
5	CARTO basemaps (cartocdn)	Map tiles (dark/light/voyager)	FREE (basemaps)	Reasonable use + attribution	Mobile FamilyMap + Web
6	Esri ArcGIS Online tiles	Satellite map tiles	Free (limited) + Paid	Limited tile usage; commercial needs subscription	Mobile/Web (SAT layer)
7	MSG91	SMS OTP	OPTIONAL (off by default) — PAID if enabled	No real free tier (~₹0.15-0.25/SMS)	Backend (optional phone OTP)
8	SMTP / Email (nodemailer)	Email OTP — PRIMARY	Free (provider tier)	Gmail ~500/day · SendGrid 100/day free	Backend (primary OTP)
9	Expo Push Notifications	Push (FCM/APNs)	FREE	Unlimited (per-second rate limit)	Mobile + Backend
10	Expo EAS Update (expo-updates)	OTA JS updates	Free tier + Paid	~1,000 update MAU/mo free	Mobile
11	Stripe	Payments	Free integration	\$0/mo; ~2.9% + \$0.30 / txn	Backend + Web
12	Razorpay	Payments (India)	Free integration	\$0/mo; ~2% / txn	Backend + Web
13	PayPal	Payments	Free integration	\$0/mo; ~2.9-3.49% + fixed / txn	Backend + Web
14	M-Pesa (Safaricom Daraja)	Mobile money (Kenya)	Free API + txn charges	Sandbox free; live per M-Pesa tariff	Backend + Web
15	PesaPal	Payments (East Africa)	Free integration	~3.5% / txn	Backend + Web
16	Sentry	Error monitoring	Free tier + Paid	5,000 errors/mo, 1 user	Mobile/Backend (referenced)
17	Google OAuth (Sign-In)	Authentication	FREE	No cost	Mobile + Web + Backend
18	Google Fonts	Web fonts	FREE	Unlimited	Web
19	Traccar	GPS tracking server	FREE (open-source) — optional self-host, NOT running	Your own server (if you set it up)	Backend (webhook stub only)
20	Redis (BullMQ/ioredis)	Job queue / cache	Free (self) / tiered	Optional; Upstash free ~10k cmd/day	Backend (optional)
21	unpkg CDN (Leaflet)	JS/CSS CDN	FREE	Fair use	Mobile FamilyMap
22	Lorem Picsum (picsum.photos)	Placeholder images	FREE	Dev/demo only	Web/dev placeholders

Totals: ~22 external services. **Google Maps is REMOVED** (replaced by free OSM/Leaflet/CARTO + Esri satellite). **Optional paid add-on: MSG91 (SMS) — off by default; email OTP is primary and free. Paid-per-transaction (no monthly fee): Stripe, Razorpay, PayPal, M-Pesa, PesaPal.** Everything else has a usable **free tier**, so there are **no mandatory monthly API costs**.

DETAILS BY CATEGORY

1) Database — Neon PostgreSQL (@neondatabase/serverless , pg)

- **Free tier:** ~0.5 GB storage, 1 project/branch, ~190 compute-hours/month (auto-suspends when idle). Good for dev + small prod.
- **Paid:** Launch ~\$19/mo (more storage/compute, no auto-suspend).
- Used by: backend (all data).

2) Object Storage — Cloudflare R2 (@aws-sdk/client-s3)

- **Free tier:** 10 GB storage/mo, 1M Class-A (write) ops, 10M Class-B (read) ops/mo, and **no egress/bandwidth fees** (R2's big advantage).
- **Paid beyond:** \$0.015/GB-month storage + op fees.
- Used by: backend avatar/circle-icon uploads (presigned URLs).

3) Maps — all FREE (OpenStreetMap + Leaflet + CARTO + Esri)

The app **and** website render maps with **Leaflet** on free tile sources — **no API key, no billing card, no monthly map bill**. - **OpenStreetMap** tiles: **Free** under the OSM tile usage policy (no heavy/bulk usage; add attribution). - **CARTO basemaps** (`basemaps.cartocdn.com` — dark/light/voyager): **Free** basemap tiles for reasonable use with attribution. - **Esri ArcGIS Online** (`server.arcgisonline.com` — satellite/SAT layer): free for limited use; commercial/high-volume needs an ArcGIS subscription. - **Google Maps Platform — REMOVED / not used**. The previously-required `EXPO_PUBLIC_GOOGLE_MAPS_API_KEY` and `react-native-maps` Google dependency are no longer needed. Maps now run entirely on the free Leaflet/OSM/CARTO stack above, eliminating the billing-account requirement.

4) SMS OTP — MSG91 — OPTIONAL (off by default)

- **Disabled by default** to avoid the only meaningful recurring paid cost. Email OTP (below) is the primary verification method.
- **If you enable it:** it's **PAID — no meaningful free tier**. Pay per SMS (~₹0.15–0.25 per transactional SMS in India; varies by country/route). Small trial credits sometimes on signup.
- With no MSG91 key set (default), the backend returns `dev_otp` / relies on email OTP instead of sending SMS — so it costs ₹0.

5) Email OTP — SMTP (nodemailer) — PRIMARY

- **Now the primary OTP channel** — free via standard SMTP free tiers (and dev mode), which is why no paid SMS is required.
- Cost depends on the SMTP provider you configure:
- **Gmail SMTP:** ~500 emails/day free (personal) — not for bulk.
- **SendGrid:** 100 emails/day free forever.
- **Brevo/Mailgun/SES:** various free tiers.
- If unset → backend logs/returns the OTP (dev mode), ₹0.

6) Push Notifications — Expo Push (expo-notifications , EXPO_ACCESS_TOKEN , exp.host)

- **FREE**. Unlimited notifications (rate-limited per second). Uses FCM (Android)/APNs (iOS) underneath, both free.

7) OTA Updates — Expo EAS Update (expo-updates)

- **Free tier:** ~1,000 monthly-active-users for OTA updates + limited bandwidth. Paid EAS plans for more.

8) Payment Gateways (5) — all free to integrate, charge per transaction

Gateway	Region	Approx fee	Sandbox
Stripe	Global	~2.9% + \$0.30/txn	Free
Razorpay	India	~2%/txn	Free
PayPal	Global	~2.9-3.49% + fixed/txn	Free (<code>api-sandbox.paypal.com</code>)
M-Pesa (Safaricom Daraja)	Kenya	per M-Pesa tariff	Free (<code>sandbox.safaricom.co.ke</code>)
PesaPal	East Africa	~3.5%/txn	Free (<code>cybqa.pesapal.com</code>)
- No monthly subscription for any — you only pay the % when money is collected.			

9) Error Monitoring — Sentry

- **Free tier:** 5,000 errors/month, 1 user, short retention. Paid Team plan ~\$26/mo for more.

10) Auth — Google OAuth / Google Sign-In

- **FREE.** No cost for authentication / "Continue with Google".

11) GPS Tracking — Traccar — optional self-host (NOT currently running) (webhook)

TRACCAR_WEBHOOK_SECRET)

- **FREE / open-source** if you choose to self-host it later. Currently the backend only has a placeholder `/webhooks/traccar` stub — no Traccar server is running and no positions are ingested from it. Phone GPS is reported by the app directly. Self-host Traccar later (its own Docker/server) if you want dedicated device tracking — no managed/per-request fees, just your server cost.

12) Queue/Cache — Redis (`bullmq` , `ioredis`) — optional

- Backend runs in node-cron mode by default (no Redis needed). If you scale: self-host Redis (free) or managed (e.g., Upstash free ~10k commands/day).

13) CDNs / Misc (all FREE)

- **unpkg** — serves Leaflet JS/CSS in the FamilyMap WebView.
- **Google Fonts** — web fonts.
- **Lorem Picsum** (`picsum.photos`) — placeholder images (development/demo only; replace for production).

✓ WHAT YOU MUST PAY FOR (to run in production)

There are **no mandatory monthly API costs**. The only spend is: 1. **Payment gateway fees** — only when you actually collect money (% per transaction, no fixed cost). 2. **Hosting/DB/storage** beyond free tiers (Neon, R2, your own VPS) as usage grows. 3. **MSG91 SMS** — *optional, off by default*. Enable only if you want real phone-OTP via SMS; otherwise the free email-OTP/dev-OTP path covers verification.

Note: **Google Maps is no longer a cost** — maps run on the free OSM/Leaflet/CARTO (+ Esri satellite) stack.

FREE FULLY FREE (within limits) — everything else

Maps (OSM/CARTO/ArcGIS via Leaflet — Google Maps removed), email OTP (provider free tier, **primary**), Expo Push, Expo Updates, Google OAuth, Google Fonts, Sentry (5k/mo), Traccar (optional self-host — not running), Redis (optional), Neon (0.5 GB), Cloudflare R2 (10 GB, \$0 egress), CDNs.

Generated for GravityPro. Verify each provider's current pricing/limits before going live.